

Module 11: Genomics and Biotechnology - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. What does PCR do?

- A. cuts DNA into random pieces
- B. reads the sequence of a protein
- C. makes many copies of a selected DNA sequence
- D. measures the pH of a sample
- Answer: C
- Why: PCR amplifies a chosen DNA region so there is enough material to analyze.

2. Gel electrophoresis separates DNA fragments by:

- A. size, with smaller fragments moving farther
- B. color of the fragments
- C. temperature alone
- D. how much protein they contain
- Answer: A
- Why: An electric field pulls DNA through the gel. Smaller fragments move faster and travel farther.

3. CRISPR is best described as a tool that can:

- A. only measure the size of a genome
- B. create energy for cells
- C. replace the whole field of biology
- D. target and edit specific DNA sequences
- Answer: D
- Why: CRISPR, adapted from a bacterial defense system, lets scientists edit DNA at chosen locations.

4. Why does biotechnology require bioethics?

- A. because the tools never actually work
- B. because powerful genetic tools raise questions of benefit, harm, and rights
- C. because DNA is not real
- D. because ethics stops all research
- Answer: B
- Why: Bioethics weighs benefits, risks, and responsibilities so that powerful genetic tools are used wisely.